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2.10.26

DBMS LAB 8- Implement Grouping and Ordering: Study and implement the GROUP BY and HAVING clauses, ORDER BY clause.

1.GROUPBY()-

```
1  CREATE TABLE employee (  
2      emp_id INT PRIMARY KEY,  
3      emp_name VARCHAR(50),  
4      email VARCHAR(100),  
5      joining_date DATE,  
6      dob DATE,  
7      salary INT  
8  );  
9  
10 INSERT INTO employee (emp_id, emp_name, email, joining_date, dob, salary) VALUES  
11 (101, 'john', 'ABC@exp.com', '2023-01-15', '1995-04-12', 65000),  
12 (102, 'smith', 'jnc@exp.com', '2022-05-20', '1990-08-25', 78000),  
13 (103, 'allision', 'MNC@exp.com', '2021-11-10', '1988-12-03', 82000),  
14 (104, 'chandler', 'YUQ@exp.com', '2023-08-01', '1998-02-14', 55000),  
15 (105, 'charlie', 'GHe@exp.com', '2020-03-12', '1992-06-30', 90000);  
16  
17 SHOW TABLES;  
18  
19 SELECT * FROM employee;  
20  
21  
22 SELECT  
23     YEAR(joining_date) AS joining_year,  
24     COUNT(emp_id) AS total_employees,  
25     SUM(salary) AS total_salary  
26 FROM employee  
27 GROUP BY YEAR(joining_date);  
28
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | email      | joining_date | dob      | salary |
+-----+-----+-----+-----+-----+-----+
| 101    | john    | ABC@exp.com | 2023-01-15   | 1995-04-12 | 65000 |
| 102    | smith   | jnc@exp.com | 2022-05-20   | 1990-08-25 | 78000 |
| 103    | allision | MNC@exp.com | 2021-11-10   | 1988-12-03 | 82000 |
| 104    | chandler | YUQ@exp.com | 2023-08-01   | 1998-02-14 | 55000 |
| 105    | charlie  | GHe@exp.com | 2020-03-12   | 1992-06-30 | 90000 |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+
| joining_year | total_employees | total_salary |
+-----+-----+-----+
| 2023 | 2 | 120000 |
| 2022 | 1 | 78000 |
| 2021 | 1 | 82000 |
| 2020 | 1 | 90000 |
+-----+-----+-----+
```

2.HAVING()-

```
1 CREATE TABLE employee (
2     emp_id INT PRIMARY KEY,
3     emp_name VARCHAR(50),
4     email VARCHAR(100),
5     joining_date DATE,
6     dob DATE,
7     salary INT
8 );
9
10 INSERT INTO employee (emp_id, emp_name, email, joining_date, dob, salary) VALUES
11 (101, 'john', 'ABC@exp.com', '2023-01-15', '1995-04-12', 65000),
12 (102, 'smith', 'jnc@exp.com', '2022-05-20', '1990-08-25', 78000),
13 (103, 'allision', 'MNC@exp.com', '2021-11-10', '1988-12-03', 82000),
14 (104, 'chandler', 'YUQ@exp.com', '2023-08-01', '1998-02-14', 55000),
15 (105, 'charlie', 'GHe@exp.com', '2020-03-12', '1992-06-30', 90000);
16
17 SHOW TABLES;
18
19 SELECT * FROM employee;
20
21
22
23 SELECT
24     YEAR(joining_date) AS joining_year,
25     AVG(salary) AS avg_salary
26 FROM employee
27 GROUP BY YEAR(joining_date)
28 HAVING AVG(salary) > 70000;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | email      | joining_date | dob      | salary |
+-----+-----+-----+-----+-----+-----+
| 101    | john    | ABC@exp.com | 2023-01-15   | 1995-04-12 | 65000 |
| 102    | smith   | jnc@exp.com | 2022-05-20   | 1990-08-25 | 78000 |
| 103    | allision | MNC@exp.com | 2021-11-10   | 1988-12-03 | 82000 |
| 104    | chandler | YUQ@exp.com | 2023-08-01   | 1998-02-14 | 55000 |
| 105    | charlie | GHe@exp.com | 2020-03-12   | 1992-06-30 | 90000 |
+-----+-----+-----+-----+-----+-----+

+-----+-----+
| joining_year | avg_salary |
+-----+-----+
| 2022         | 78000.0000 |
| 2021         | 82000.0000 |
| 2020         | 90000.0000 |
+-----+-----+
```

3.ORDERBY()-

```
1 CREATE TABLE employee (
2     emp_id INT PRIMARY KEY,
3     emp_name VARCHAR(50),
4     email VARCHAR(100),
5     joining_date DATE,
6     dob DATE,
7     salary INT
8 );
9
10 INSERT INTO employee (emp_id, emp_name, email, joining_date, dob, salary) VALUES
11 (101, 'john', 'ABC@exp.com', '2023-01-15', '1995-04-12', 65000),
12 (102, 'smith', 'jnc@exp.com', '2022-05-20', '1990-08-25', 78000),
13 (103, 'allision', 'MNC@exp.com', '2021-11-10', '1988-12-03', 82000),
14 (104, 'chandler', 'YUQ@exp.com', '2023-08-01', '1998-02-14', 55000),
15 (105, 'charlie', 'GHe@exp.com', '2020-03-12', '1992-06-30', 90000);
16
17 SHOW TABLES;
18
19 SELECT * FROM employee;
20
21 SELECT
22     emp_id,
23     emp_name,
24     salary
25 FROM employee
26 ORDER BY salary DESC;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | email      | joining_date | dob      | salary |
+-----+-----+-----+-----+-----+-----+
| 101    | john    | ABC@exp.com | 2023-01-15   | 1995-04-12 | 65000 |
| 102    | smith   | jNC@exp.com | 2022-05-20   | 1990-08-25 | 78000 |
| 103    | allision | MNC@exp.com | 2021-11-10   | 1988-12-03 | 82000 |
| 104    | chandler | YUQ@exp.com | 2023-08-01   | 1998-02-14 | 55000 |
| 105    | charlie  | GHe@exp.com | 2020-03-12   | 1992-06-30 | 90000 |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+
| emp_id | emp_name | salary |
+-----+-----+-----+
| 105    | charlie  | 90000 |
| 103    | allision | 82000 |
| 102    | smith   | 78000 |
| 101    | john    | 65000 |
| 104    | chandler | 55000 |
+-----+-----+-----+
```
